Calculas:

L. Hopital's Rule:

Indeaterminate Form →

(i) $\frac{0}{0}$

(ii) $\frac{\infty}{\infty}$

(iii) $0.\infty$

(iv) 0^0

(v) ∞^0

(vi) 1^∞

(vii) $\infty - \infty$

Problem: $\left(\frac{0}{0}\text{ Form}\right)$

(1) $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$ $\left[\frac{0}{0}\text{ Form}\right]$

By L. Hopital's Rule,

$$\lim_{x \to 3} \frac{\frac{d}{dx}(x^2 - 9)}{\frac{d}{dx}(x - 3)}$$

$$\Rightarrow \lim_{x \to 3} \left(\frac{2x}{1}\right)$$

$$\Rightarrow \frac{(2 \times 3)}{1} = 6$$

(Ans)

(ii) $\lim_{x \to 2} \frac{x^7 - 128}{x^3 - 8}$ $\left[\frac{0}{0} \text{ form}\right]$

By L. Hopital's Rule.

$$\lim_{x \to 2} \frac{\frac{d}{dx}(x^7 - 128)}{\frac{d}{dx}(x^3 - 8)}$$

$$= \lim_{x \to 2} \left(\frac{7x^6}{3x^2}\right)$$

$$= \frac{7 \times 2^6}{3 \times 2^2}$$

$$= \frac{448}{12} = 37.33$$

(Ans)

(iii) $\lim_{x \to 0} \frac{x - \sin x}{x^3}$ $\left[\frac{0}{0} \text{ form}\right]$

By L. Hopital's Rule.

$$\lim_{x \to 0} \frac{\frac{d}{dx}(x - \sin x)}{\frac{d}{dx}(x^3)}$$

$\left[\begin{array}{l} * \frac{d}{dx} \cos x = -\sin x \\ * \frac{d}{dx} \sin x = \cos x \end{array}\right]$

$$\Rightarrow \lim_{x \to 0} \frac{(1 - \cos x)}{3x^2}$$ $\left[\frac{0}{0} \text{ form}\right]$

$$= \lim_{x \to 0} - \left(\frac{-\sin x}{3.2x}\right)$$

$$= \lim_{x \to 0} \frac{\sin x}{6x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \frac{\cos 0}{6}$$

$$= 1/6$$

(Ans)

$\left[\frac{\infty}{\infty} \text{ Form}\right]$

(iv) $$\lim_{x \to \infty} \frac{2x^2 - 3x + 1}{3x^2 + 5x - 2}$$

By L. Hopital's Rule,

$$= \lim_{x \to \infty} \frac{4x - 3}{6x + 5} \quad \left[\frac{\infty}{\infty} \text{ Form}\right]$$

$$= \lim_{x \to \infty} \frac{4}{6}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}$$

(Ans)

(v) $\lim_{x \to 0} \frac{xe^x - \log(1+x)}{x^2}$ $\left[\frac{0}{0} \text{ form}\right]$

By L. Hopital's law,

$\lim_{x \to 0} \frac{xe^x + e^x - \frac{1}{1+x}}{2x}$

$= \lim_{x \to 0} \frac{xe^x + e^x + e^x + \frac{1}{(1+x)^2}}{2}$

$= \frac{0+1+1+1}{2}$

$= \frac{3}{2}$

(Ans)

$\frac{d}{dx} uv = u \frac{d}{dx}(v) + v \frac{d}{dx}(u)$

$= x \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(x)$

$= xe^x + e^x . 1$

$= xe^x + e^x$

04.11.2024

① Problem : Evaluate $\lim_{x \to 0} (\cos x)^{\cot x} = ?$

Solve :

$$\text{let } y = \lim_{x \to 0} (\cos x)^{\cot x} \qquad [1^{\infty} \text{ Form}]$$

$$\Rightarrow \ln y = \lim_{x \to 0} \ln (\cos x)^{\cot x}$$

$$\Rightarrow \ln y = \lim_{x \to 0} \cot x \ln(\cos x)$$

$$\Rightarrow \ln y = \lim_{x \to 0} \frac{\ln \cos x}{\tan x} \qquad \left[\frac{0}{0} \text{ Form}\right] \quad \left[\because \cot x = \frac{1}{\tan x}\right]$$

By L. Hopital's law.

$$\ln y = \lim_{x \to 0} \frac{\frac{1}{\cos x} \times (-\sin x)}{\sec^2 x}$$

$$= \lim_{x \to 0} \left(\frac{-\sin x}{\cos x . \sec^2 x}\right)$$

$$= \lim_{x \to 0} \left(\frac{-\sin x}{\cos x . \frac{1}{\cos 2x}}\right)$$

$$= \lim_{x \to 0} \left(\frac{-\sin x}{\frac{1}{\cos x}}\right)$$

$= \lim_{x \to 0} -\sin x \cdot \cos x$

$= -\sin 0 \cdot \cos 0$

$= -0 \cdot \underline{1}$

$= 0$

$\Rightarrow \ln y = 0$

$\Rightarrow y = e^0$

$\Rightarrow y = 1$

$\therefore \lim_{x \to 0} (\cos x)^{\cot x} = \underline{1}$

(Ans)

(ii) Problem: Evaluate $\lim_{x \to 0} \frac{e^x - e^{\sin x}}{x - \sin x}$ $\left[\frac{0}{0} \text{ Form}\right]$

Solve: By L. Hopital's Rule.

$$= \lim_{x \to 0} \frac{e^x - e^{\sin x}(\cos x)}{1 - \cos x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x \to 0} \frac{e^x - \left[e^{\sin x} \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(e^{\sin x})\right]}{0 - (-\sin x)}$$

$$= \lim_{x \to 0} \frac{e^x - \left[e^{\sin x}(-\sin x) + \cos x \cdot e^{\sin x} \cdot \cos x\right]}{\sin x}$$

$$= \lim_{x \to 0} \frac{e^x + \sin x \cdot e^{\sin x} - \cos^2 x \cdot e^{\sin x}}{\sin x}$$

$$= \lim_{x \to 0} \frac{(e^x - \cos^2 x \cdot e^{\sin x}) + \sin x \cdot e^{\sin x}}{\sin x}$$

$$= \lim_{x \to 0} \left(\frac{e^x - \cos^2 x \cdot e^{\sin x}}{\sin x}\right) + \lim_{x \to 0} \frac{\sin x \cdot e^{\sin x}}{\sin x}$$

$$= \lim_{x \to 0} \frac{e^x - \cos^2 x \cdot e^{\sin x}}{\sin x} + \lim_{x \to 0} (e^{\sin x})$$

$$= \lim_{x \to 0} \frac{e^x - \left[\cos^2 x \cdot e^{\sin x} \cdot \cos x + e^{\sin x} \cdot 2\cos x \cdot \sin x\right]}{\cos x} + 1$$

$$= \frac{1 - [1 - 0]}{1} + 1$$

$$= \frac{1 - 1}{1} + 1$$

$$= \frac{0}{1} + 1$$

$$= 1$$

(Ans)

(iii) Problem: Evaluate $\lim_{x \to 0} \left(\frac{1}{x} - \cot x\right)$

Solve:

Given,

$$\lim_{x \to 0} \left(\frac{1}{x} - \cot x\right) \quad \boxed{\infty - \infty \text{ Form}}$$

$$= \lim_{x \to 0} \left(\frac{1}{x} - \frac{\cos x}{\sin x}\right)$$

$$= \lim_{x \to 0} \frac{\sin x - x\cos x}{x \sin x} \quad \boxed{\frac{0}{0} \text{ Form}}$$

By L. Hopital's Rules.

$$\lim_{x\to 0} \frac{\cos x - \left[x \frac{d}{dx}\cos x + \cos x \frac{d}{dx}(x)\right]}{x\frac{d}{dx}\sin x + \sin x \frac{d}{dx}(x)}$$

$$= \lim_{x\to 0} \frac{\cos x - [x(-\sin x) + \cos x]}{x(\cos x) + \sin x}$$

$$= \lim_{x\to 0} \frac{\cos x + x\sin x - \cos x}{x\cos x + \sin x}$$

$$= \lim_{x\to 0} \frac{x\sin x}{x\cos x + \sin x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x\to 0} \frac{x\frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(x)}{x\frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(x) + \cos x}$$

$$= \lim_{x\to 0} \frac{x\cos x + \sin x}{-x\sin x + 2\cos x}$$

$$= \lim_{x\to 0} \frac{x\cos x + \sin x}{2\cos x - x\sin x}$$

$$= \frac{0+0}{2-0}$$

$$= 0$$

(Ans)

(iv) Evaluate. $\lim_{x \to \pi/2} (\sin x)^{\tan x}$

Let, $y = \lim_{x \to \pi/2} (\sin x)^{\tan x}$ $[1^\infty \text{ Form}]$

$\Rightarrow \ln y = \lim_{x \to \pi/2} \ln (\sin x)^{\tan x}$

$\Rightarrow \ln y = \lim_{x \to \pi/2} \tan x \ln (\sin x)$

$\Rightarrow \ln y = \lim_{x \to \pi/2} \frac{\ln \sin x}{\cot x}$ $\left[\frac{0}{0} \text{ Form}\right]$

By L. Hopital's Rules.

$$\ln y = \lim_{x \to \pi/2} \frac{\frac{1}{\sin x} \cdot \cos x}{-\operatorname{cosec}^2 x}$$

$$= \lim_{x \to \pi/2} \frac{\cos x}{\sin x \cdot (-\operatorname{cosec}^2 x)}$$

$$= \lim_{x \to \pi/2} \frac{\cos x}{\sin x \cdot \left(-\frac{1}{\sin^2 x}\right)}$$

$$= \lim_{x \to \pi/2} \frac{\cos x}{-\frac{1}{\sin x}}$$

$$= \lim_{x \to \pi/2} \cos x \cdot (-\sin x)$$

$$= \cos \pi/2 \cdot (-\sin \pi/2)$$

$$= 0 \cdot (-1)$$

$$= 0$$

$\Rightarrow \ln y = 0$

$\Rightarrow y = e^0$

$\Rightarrow y = 1$

$$\therefore \lim_{x \to \pi/2} (\sin x)^{\tan x} = \underline{1}$$

(Ans)

7.11.202

(i) Evaluate, $\lim_{x \to 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$

(ii) Evaluate, $\lim_{x \to 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

(iii) Evaluate, $\lim_{x \to 0} \frac{a^x - 1 - x \ln a}{x^2}$

(iv) Evaluate, $\lim_{x \to 0} \frac{e^x - e^{-x} - 2\ln(1+x)}{x \sin x}$

(v) Evaluate, $\lim_{x \to 0} \frac{3\tan x - 3x - x^3}{x^5}$

(vi) Evaluate, $\lim_{x \to 0} (\sin x)^x$

(vii) Evaluate, $\lim_{x \to 0} (\cos x)^{\cot^2 x}$

① Given.

$$\lim_{x \to 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right) \qquad [\infty - \infty \text{ Form}]$$

$$= \lim_{x \to 0} \left(\frac{\sin^2 x - x^2}{x^2 \sin^2 x}\right) \qquad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x \to 0} \left[\frac{\left(x - \left(\frac{1}{6}\right)x^3 + \cdots\right)^2 - x^2}{x^2\left(x - \left(\frac{1}{6}\right)x^3 + \cdots\right)^2}\right] \qquad \left[\because \sin x = x - \frac{x^3}{3!} + \cdots\right]$$

$$= \lim_{x \to 0} \left[\frac{x^2 - \left(\frac{1}{3}\right)x^4 + \cdots\Big) - x^2}{x^2\left(x^2 - \left(\frac{1}{3}\right)x^4 + \cdots\right)}\right] \qquad \left[(a-b+\cdots)^2 = a^2 - 2ab + \cdots\right]$$

$$= \lim_{x \to 0} \left[\frac{\left(-\frac{1}{3}\right)x^4 + \text{Higher power of } x}{x^4 - \frac{1}{3}x^6 + \cdots}\right]$$

$$\frac{-\frac{1}{3} + 0 + \cdots}{1 - 0 + \cdots}$$

$$\frac{1}{3}$$

(Ans)

(2) Given,

$$\lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x\to 0} \frac{e^x + e^{-x} - 2}{1 - \cos x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x\to 0} \frac{e^x - e^{-x}}{\sin x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x\to 0} \frac{e^x + e^{-x}}{\cos x}$$

$$= \frac{e^0 + e^{-0}}{\cos 0}$$

$$= \frac{1+1}{1} = 2$$

(Ans)

(iii) Given,

$$\lim_{x\to 0} \frac{a^x - 1 - x\ln a}{x^2} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$\Rightarrow \lim_{x\to 0} \frac{a^x \ln a - \ln a}{2x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x\to 0} \frac{a^x (\ln a)^2 - 0}{2}$$

$$= \lim_{x\to 0} \frac{a^0 (\ln a)^2}{2}$$

$$= \frac{1}{2} (\ln a)^2$$

(Ans)

(4) Given,

$$\lim_{x\to 0} \frac{e^x - e^{-x} - 2\ln(1+x)}{x\sin x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x\to 0} \frac{e^x + e^{-x} - 2(1+x)^{-1}}{x\cos x + \sin x} \quad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x\to 0} \frac{e^x - e^{-x} + 2(1+x)^{-2}}{\cos x - x\sin x + \cos x}$$

$$= \frac{1-1+2}{1-0+1} = 1$$

(Ans)

(5) Given.

$$\lim_{x \to 0} \frac{3\tan x - 3x - x^3}{x^5} \qquad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x \to 0} \frac{3\sec^2 x - 3 - 3x^2}{5x^4}$$

$$= \lim_{x \to 0} \frac{6\sec^2 x \cdot \tan x - 6x}{20x^3} \qquad \left[\frac{0}{0} \text{ Form}\right]$$

$$= \lim_{x \to 0} \frac{12\sec^2 \cdot x \tan x + 6\sec^4 x - 6}{60x^2}$$

$$= \lim_{x \to 0} \frac{18\sec^4 x - 12\sec^3 x + 6}{60x^2} \qquad \left[\frac{0}{0} \text{ Form}\right] \quad \left[\sec^2 x - 1 = \tan^2 x\right]$$

$$= \lim_{x \to 0} \frac{72\sec^4 \cdot \tan x - 24\sec^2 x - 6}{120x}$$

$$= \lim_{x \to 0} \frac{288\sec^4 \cdot x \tan^2 x + 72\sec^6 x - 48\sec^2 \cdot x \tan^2 x - 24\sec^4 x}{120}$$

$$= \frac{0 + 72 - 0 - 24}{120}$$

$$= \frac{48}{120} = 2/5$$

(Ans)

(6) Given,

$$\lim_{x\to 0} (\sin x)^x$$

Let, $y = \lim_{x\to 0} (\sin x)^x$

$\Rightarrow \ln y = \lim_{x\to 0} x \ln \sin x$

$= \lim_{x\to 0} \frac{\ln \sin x}{1/x}$

$= \lim_{x\to 0} \frac{\frac{1}{\sin x} \cdot \cos x}{-1/x^2}$

$= \lim_{x\to 0} \left(\frac{-x^2}{\tan x}\right)$

$= -\lim_{x\to 0} \frac{2x}{\sec^2 x}$

$= -\frac{2\times 0}{\sec^2 0}$

$= 0$

$\Rightarrow \ln y = 0$

$\Rightarrow y = e^0$

$\Rightarrow y = 1$

$\therefore \lim_{x\to 0} (\sin x)^x = 1$ (Ans)

(7) Let,

$$y = \lim_{x \to 0} (\cos x)^{\cot^2 x} \qquad [1^\infty \text{ form}]$$

$$\Rightarrow \ln y = \cot^2 x \ln(\cos x)$$

$$= \lim_{x \to 0} \frac{\ln(\cos x)}{\tan^2 x} \qquad \left[\frac{0}{0} \text{ form}\right]$$

$$= \lim_{x \to 0} \frac{\frac{1}{\cos x} \cdot (-\sin x)}{2\tan x \cdot \sec^2 x}$$

$$= \lim_{x \to 0} \frac{-\cancel{\tan x}}{2\cancel{\tan x} \cdot \sec^2 x}$$

$$= -1/2 \lim_{x \to 0} \cos^2 x$$

$$= -1/2 (\cos^2 0) = -1/2$$

$$\Rightarrow \ln y = -1/2$$

$$\Rightarrow y = e^{-1/2}$$

$$\Rightarrow y = \frac{1}{\sqrt{e}}$$

$$\therefore \lim_{x \to 0} (\cos x)^{\cot^2 x} = \frac{1}{\sqrt{e}}$$

(Ans)

11.11.2024

Continuity :- A function $F(x)$ is said to be continuous at $x=a$ if;

$$\lim_{x \to a^+} F(x) = \lim_{x \to a^-} F(x) = F(a)$$

R.H.L. L.H.L

$F(x) \to$ functional value at $x=a$

Problem :- ① A function $F(x)$ is defined as follows.

$$F(x) = \begin{cases} 5x-4 & \text{when } 0 < x \leq 1 \\ 4x^2-3x & \text{when } 1 < x < 2 \\ 3x+4 & \text{when } x \geq 2 \end{cases}$$

Find wheather the function in continuous at $x=1$ and $x=2$

Solve :-

A.1 , $x=1$:-

When, $x=1$ then $F(x) = 5x-4$

$= 5.1-4$

$= 1$

R.H.S. $\rightarrow \lim_{x \to 1^+} F(x) = \lim_{x \to 1^+} (4x^2 - 3x)$

$= 4.1^2 - 3.1$

$= 1$

L.H.S. $\rightarrow \lim_{x \to 1^-} F(x) = \lim_{x \to 1^-} (5x - 4)$

$= 5.1 - 4$

$= 1$

Since, $\lim_{x \to 1^+} F(x) = \lim_{x \to 1^-} F(x) = F(1)$

Therefore, The given Function in Continuous at $x = 1$.

(Ans)

A.2 , $x = 2$;

When, $x = 2$ then $F(x) = 3x + 4$

$= 3.2 + 4$

$= 10$

$\lim_{x \to 2^+} F(x) = \lim_{x \to 2^+} (3x + 4)$

$= 3.2 + 4$

$= 10$

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (4x^2 - 3x)$$

$$= 4 \cdot 2^2 - 3 \cdot 2$$

$$= 16 - 6$$

$$= 10$$

Since, $\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x) = f(2)$

Therefore, The given function is continuos at $x = 2$.

(Ans)

③ $F(x) = \begin{cases} 5x-4 & ; \quad 0 < x \le 1 \\ 4x^2-3x & ; \quad 1 < x < 2 \end{cases}$

A. 1 . $x = 1$;

When, $x = 1$ then $F(x) = 5x-4$

$= 5.1-4$

$= 1$

R.H.S. →

$\lim_{x \to 1^+} F(x) = \lim_{x \to 1^+} (4x^2-3x)$

$= (4.1^2-3.1)$

$= 4-3$

$= 1$

L.H.S. →

$\lim_{x \to 1^-} F(x) = \lim_{x \to 1^-} (5x-4)$

$= (5.1-4)$

$= 1$

Since, $\lim_{x \to 1^+} F(x) = \lim_{x \to 1^-} F(x) = F(1)$

Therefore, The given function is continuous at $x = 1$

(Ans)

(4)

$$F(x) = \begin{cases} x^2 & \text{when} \quad x < 0 \\ x & \text{when} \quad 0 \le x \le 1 \\ \frac{1}{x} & \text{when} \quad x > 1 \end{cases}$$

A. 1, $x = 0$;

When, $x = 0$ then $F(x) = x^2$

$= 0$

R.H.S. $\rightarrow \lim\limits_{x \to 0^+} F(x) = \lim\limits_{x \to 0^+} (x)$

$= 0$

L.H.S. $\rightarrow \lim\limits_{x \to 0^-} F(x) = \lim\limits_{x \to 0^-} (x^2)$

$= 0$

Since, $\lim\limits_{x \to 0^+} F(x) = \lim\limits_{x \to 0^-} F(x) = F(0)$

Therefore, The given function is continuous at $x = 0$

A.2. $x = 1$;

When, $x = 1$ then $f(x) = x$

$= 1$

R.H.S. $\rightarrow \lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \left(\frac{1}{x}\right)$

$= \frac{1}{1}$

$= 1$

L.H.S. $\rightarrow \lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (x)$

$= 1$

Since, $\lim_{x \to 1^+} f(x) = \lim_{x \to 1^-} f(x) = f(1)$

Therefore, the function is given continuos at $x = 1$

(Ans)

(5)

$$f(x) = \begin{cases} 3+2x & \text{when } -\frac{3}{2} \leq x \leq 0 \\ 3-2x & \text{when } 0 \leq x \leq \frac{3}{2} \\ -3-2x & \text{when } x \geq \frac{3}{2} \end{cases}$$

A.1. $x = 0$;

When, $x = 0$. then $f(x) = 3+2x$

$= 3+2\cdot 0$

$= 3$

R.H.S. $\rightarrow \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (3-2x)$

$= 3-2.0$

$= 3$

L.H.S. $\rightarrow \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (3+2x)$

$= 3+2.0$

$= 3$

Since, $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x) = f(0)$

Therefore, The given Function is continuous at $x = 0$

(Ans)

A. 2. $x = \frac{3}{2}$,

When, $x = 3/2$ then $F(x) = 3 - 2x$

$= 3 - 2 \cdot \frac{3}{2}$

$= 0$

R.H.S. $\rightarrow \lim_{x \to 3/2^+} F(x) = \lim_{x \to 3/2^+} (-3 - 2x)$

$= -3 - 2 \cdot \frac{3}{2}$

$= -3 - 3$

$= -6$

L.H.S. $\rightarrow \lim_{x \to 3/2^-} F(x) = \lim_{x \to 3/2^-} (3 - 2x)$

$= 3 - 2 \cdot \frac{3}{2}$

$= 3 - 3$

$= 0$

Since, $\lim_{x \to 3/2^+} F(x) \neq \lim_{x \to 3/2^-} F(x) = F(\frac{3}{2})$

Therefore, The given function is discontinuous at $x = \frac{3}{2}$.

(Ans)

(6)

$$F(x) \begin{cases} -x & \text{when } -2 \le x < 0 \\ x & \text{when } 0 < x < 1 \\ 3-x & \text{when } 1 \le x \le 2 \end{cases}$$

A.1. $x = 1$;

When, $x = 1$ then $F(x) = (3-x)$

$= 3-1$

$= 2$

R.H.S. $\rightarrow \lim_{x \to 1^+} F(x) = \lim_{x \to 1^+} (3-x)$

$= 3-1$

$= 2$

L.H.S. $\rightarrow \lim_{x \to 1^-} F(x) = \lim_{x \to 1^-} (x)$

$= 1$

Since. $\lim_{x \to 1^+} F(x) = \lim_{x \to 1^-} F(x) \neq F(1)$

Therefore, The given Function is discontinuous at $x = 1$

(Ans)

(7)

$$F(x) = \begin{cases} -x & \text{when } x < 0 \\ x & \text{when } 0 \le x \le 1 \\ 2-x & \text{when } x > 1 \end{cases}$$

A. 1 , $x = 1$;

When $x = 1$ then $F(x) = x$

$= 1$

R.H.L. $\rightarrow \lim_{x \to 1^+} F(x) = \lim_{x \to 1^+} (2-x)$

$= (2-1)$

$= 1$

L.H.L. $\rightarrow \lim_{x \to 1^-} F(x) = \lim_{x \to 1^-} (x)$

$= 1$

Since, $\lim_{x \to 1^+} F(x) = \lim_{x \to 1^-} F(x) = F(1)$

Therefore, the function is given continuous at $x = 1$

A.2, $x = 0$;

When $x = 0$ then $F(x) = x$

$= 0$

R.H.L. $\rightarrow \lim_{x \to 0^+} F(x) = \lim_{x \to 0^+} (x)$

$= 0$

L.H.L. $\rightarrow \lim_{x \to 0^-} F(x) = \lim_{x \to 0^-} (-x)$

$= 0$

Since, $\lim_{x \to 0^+} F(x) = \lim_{x \to 0^-} F(x) = F(0)$

Therefore, The given function is continuas at $x = 0$

(Ans)

⑧

$$F(x) = \begin{cases} 1 & \text{when } x < 0 \\ 1+\sin x & \text{when } 0 \leq x \leq \pi/2 \\ 2+(x-\pi/2)^2 & \text{when } x > \pi/2 \end{cases}$$

A.1. $x = \pi/2$;

When, $x = \pi/2$ then $F(x) = 2+(x-\pi/2)^2$

$= 2+(\pi/2 - \pi/2)^2$

$= 2$

R.H.L. $\rightarrow \lim\limits_{x \to \pi/2^+} F(x) = \lim\limits_{x \to \pi/2^+} \{2+(x-\pi/2)^2\}$

$= 2+(\pi/2 - \pi/2)^2$

$= 2$

L.H.L. $\rightarrow \lim\limits_{x \to \pi/2^-} F(x) = \lim\limits_{x \to \pi/2^-} (1+\sin x)$

$= (1+\sin \pi/2)$

$= 1+1$

$= 2$

Since. $\lim\limits_{x \to \pi/2^+} F(x) = \lim\limits_{x \to \pi/2^-} F(x) = F(\pi/2)$

Therefore. The given function is continuous at $x = \pi/2$

(Ans)